Exact VUS Evaluation by Support Events and Shared Interval Geometry

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ABSTRACT

Volume Under the Surface (VUS) evaluates anomaly scores across thresholds and temporal tolerance widths. We derive an exact support-event representation for the original dynamic VUS operator. Thresholds with no active support have zero precision–recall contribution; their receiver-operating-characteristic contribution requires a predecessor correction. Complete global ties and merged anomaly components remain explicit. Standard fixed-width ranking and two-distance interval geometry give $O(n + (L + 1)Q)$ evaluation, where Q is maximum-buffer support. Shared-label preparation further reduces repeated work across score columns. A prospectively fixed study uses 73 complete real series from three public collections and eight fixed detector procedures, producing 584 unrounded score vectors per host. Fresh execution on Apple M1 Pro and AMD EPYC preserves the main result: directly timed collection workflows are 16.2–23.0× faster than an unchanged exhaustive PR comparator. Matched ROC+PR workflows improve by 22.4–42.5× over a public cumulative-sum implementation. Compiled controls distinguish compression, implementation and sharing effects: an optimized conventional direct-ramp control is faster than our single-model implementation at the primary width, while batch evaluation improves over it. Exact evaluation also dominates the measured 250-threshold approximation in both latency and numerical fidelity on this workload. A public nine-metric consumer illustrates the scope limit: its VUS stage improves 66.1×, but total latency only 1.031× because other metrics dominate. We report paired uncertainty, shared-label dependence, unfavorable regimes and process-memory costs. Mathematical exactness and measured floating-point parity are distinct guarantees.

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The source code, data, and/or other artifacts have been made available at https://github.com/mandu5/sparse-vus/tree/research/pvldb-20261024.

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1 INTRODUCTION

Evaluation is a repeated data-analysis operation in time-series anomaly benchmarks: stored score vectors from several detectors or configurations are joined with the same labels, evaluated, and compared to select a numerical winner. Volume Under the Surface (VUS) aggregates range-aware precision–recall (PR) or receiver-operating-characteristic (ROC) integrals over temporal widths [2, 15]. Exhaustive evaluation integrates every complete distinct-score threshold at every integer width. Sampling thresholds reduces work but changes the returned scalar and can change the selected model. The computational question is whether the exact operator needs all those states.

Substantial acceleration already exists. Boniol et al. separate maximum-buffer support from unaffected regions and reuse static work. Klein's fast-VUS uses support-score cumulative sums; Benchopt uses dense cumulative sums and segment maxima [1, 2, 7]. However, reusing support statistics still leaves a second question: which threshold states must be retained to compute the scalar integral? PR and ROC give different answers, because a threshold with no new true-positive mass can still contribute ROC area.

We derive a scalar event representation of the original dynamic operator. Inactive threshold runs have zero PR area, and their ROC area is a horizontal rectangle preceding the next active transition. This distinction matters: connecting only retained ROC endpoints gives an incorrect answer, even without buffers. Complete global score ties must include predictions outside support, and expanded anomaly components must merge under the target's overlap rule. A two-distance lemma removes repeated interval-ramp construction. With standard binary64 radix preparation, exhaustive evaluation takes $O(n + (L + 1)Q)$, where n is series length, L is maximum width, and Q is maximum-width support. This is a support-sensitive bound, not a claim that sorting arbitrary real numbers is linear.

The data-management contribution is an exact implementation of a recurring analytics operator on stored detector profiles. It reduces the states and width-dependent work needed for the same aggregate, and exposes shared-label preparation when several score columns are evaluated together. The latter uses standard caching and loop interchange; its value is measured separately from the event representation. Our complete workflow reads stored arrays, evaluates all score columns, selects winners, and writes the decision table. We do not claim a new detector, deployment selector, metric, or database execution engine.

A preliminary evaluator and working manuscript were publicly released as Sparse VUS 0.3.0 on 21 October 2026 [9]. This paper

presents the complete exact-evaluator work and separately measures the sharing increment over that unrefereed software precursor.

The paper makes three connected contributions. First, it proves the support-event identities and their complexity, including the dynamic denominator, complete ties, and ROC predecessor correction. Second, it gives an implementation and preparation/streaming ablations for repeated evaluation with a shared label sequence. Third, it evaluates full public calls and complete collection workflows on a prospectively frozen, full-precision panel, followed by a separately frozen strongest-comparator and approximation study. The experiments report confidence intervals, label-block sensitivity, exact-value checks and unsuccessful alternatives. The primary study uses three public univariate collections and eight fixed score-generation procedures, with fresh execution on two CPU architectures. Record reuse and the scope of warm stored-score evaluation are stated with the experimental protocol.

2 TARGET AND EXACT EVENT REPRESENTATION

Let $y \in \{0,1\}^n$ contain both classes and $s \in \mathbb{R}^n$ contain finite scores after binary64 conversion. Larger scores indicate greater anomaly evidence. Let $P_0 = \sum_i y_i$, and let the K maximal original anomaly intervals have inclusive endpoints $[a_k, b_k]$ in temporal order. Widths are the integers $w = 0, \dots, L$. The continuous label $a_i^{(w)}$ is one on original positives. An interval contributes $\sqrt{1 - d/w}$ outside its endpoints at integer distance $1 \le d \le \lfloor w/2 \rfloor$; contributions are summed and clipped at one. At $w = 0$ no ramp is added. Expanded intervals that overlap merge, while adjacent nonoverlapping intervals remain separate. This is the original dynamic VUS target pinned to the official `RangeAUC_volume_opt` source at revision `467bace0`, configured for exhaustive thresholds [21]. It is not shared by all software named VUS.

For a threshold t, define

$$p_i(t) = \mathbf{1}\{s_i \ge t\}, \qquad C(t) = \sum_i p_i(t), \tag{1}$$

$$A(t) = \sum_i y_i p_i(t), \qquad B(t) = \sum_i (1 - y_i) a_i^{(w)} p_i(t), \tag{2}$$

$$T(t) = A(t) + B(t), \qquad P(t) = P_0 + B(t)/2, \tag{3}$$

$$N(t) = n - P(t). \tag{4}$$

Let $H(t)$ count expanded components hit by at least one predicted positive, and let K_w count those components. The curve coordinates are

$$R(t) = \min\{T(t)/P(t), 1\}\, H(t)/K_w, \tag{5}$$

$$F(t) = (C(t) - T(t))/N(t), \qquad \pi(t) = T(t)/C(t). \tag{6}$$

Precision is used only at nonempty retained thresholds, where $C(t) > 0$; the initial origin requires no precision value. The denominator is positive: $P(t) \le (n + P_0)/2 < n$. ROC integrates trapezoids from the origin through complete distinct descending score thresholds to the explicit terminal $(1, 1)$. PR uses the right-step sum $\sum_j (R_j - R_{j-1})\pi_j$, with $R_0 = 0$. Each VUS scalar is the arithmetic mean over every width. Signed horizontal ROC increments are preserved; the derivation does not replace a dynamic denominator by a fixed one.

Let $S_w = \{i : a_i^{(w)} > 0\}$, $Q_w = |S_w|$, and $Q = Q_L$. For an active event with score v_j, precompute the complete global counts $c_j^{>} = |\{i : s_i > v_j\}|$ and $c_j^{\ge} = |\{i : s_i \ge v_j\}|$. They include positions outside S_w and outside S_L.

Lemma 1 (Inactive threshold runs). *Between consecutive active score events at a fixed width, T, B, P, N, H, R are constant. Omitted thresholds contribute a horizontal ROC rectangle and zero PR area. At an active score, its complete global numerical tie is a single transition.*

Proof. A newly admitted inactive point has zero weight and cannot hit an expanded component. Only C changes. The horizontal increments $(\Delta C)/N$ telescope at fixed height R. All positions tied with an active event enter at its global threshold, so both counts must retain them. □

If the previous retained state is (T_o, N_o, R_o, c_o), event j contributes

$$f_j^{-} = (c_j^{>} - T_o)/N_o, \tag{7}$$

$$\Delta\mathrm{ROC}_j = (c_j^{>} - c_o)R_o/N_o + (F_j - f_j^{-})(R_j + R_o)/2, \tag{8}$$

$$\Delta\mathrm{PR}_j = (R_j - R_o)T_j/c_j^{\ge}. \tag{9}$$

The initial state is $(0, n - P_0, 0, 0)$, and afterwards $c_o = c_j^{\ge}$. Connecting retained endpoints without the intervening rectangle is incorrect. For example, $y = (1, 0, 1, 0)$, $s = (3, 2, 1, 0)$ and $L = 0$ give ROC area 0.625; connecting support endpoints alone gives 0.8125.

Lemma 2 (Two-distance geometry). *The weight of a nonoriginal point is determined by its distances d_1, d_2 to the nearest two distinct original intervals. It is zero if $d_1 > \lfloor w/2 \rfloor$, one if $d_2 \le \lfloor w/2 \rfloor$, and $\sqrt{1 - d_1/w}$ otherwise.*

Proof. Every active ramp has weight at least $1/\sqrt{2}$. Two contributions therefore clip to one. If exactly one interval contributes, its distance determines the weight. Distances to ordered intervals are monotone moving away on each side, so two neighbors on each side contain the nearest two intervals. □

Theorem 1 (Support-event evaluation). *In a unit-cost model with score comparisons and metric arithmetic, including square roots, both exhaustive scalar integrals can be computed in $O(n \log(Q+1) + (L+1)Q)$ time and $O(n + Q)$ auxiliary words.*

Proof. Extract original intervals and maximum support in $O(n)$ time. Sort the Q support scores and group equal values. Binary-search all n scores among these groups to obtain complete global counts $c^{>}, c^{\ge}$ without sorting off-support scores, at cost $O(n \log(Q + 1))$. The ordered-interval geometry can be prepared within this bound. At each width, compute support weights, component membership, grouped masses and first component-hit events in $O(Q + K)$ time. A scan of the at most Q event groups integrates the preceding rectangles, complete tied transitions and terminal ROC tail. The inactive-run lemma makes this identical to the exhaustive target in exact metric arithmetic. Stream the

width mean using input-, support- and interval-sized arrays. Since $K \le Q \le n$, the stated bounds follow. □

Proposition 1 (Fixed-binary64 preparation). *For finite binary64 scores, complete global tie counts, the score ordering of maximum support, and its two-distance geometry can be prepared in $O(n + Q + K)$ time and $O(n + Q)$ auxiliary words. Discrete work uses a word-RAM with input indices, counts, widths, and intermediate index arithmetic fitting words. Metric arithmetic, including square roots, has unit cost.*

Proof. Normalize signed zeros. Complement a negative score's bit pattern and flip the sign bit of a nonnegative score, then complement the result to obtain a descending numerical key. Eight stable byte-counting passes order these 64-bit keys in $O(n)$ time. Scan complete ties and retain their starting and ending global ranks. Filter the permutation to maximum-support positions. A temporal scan with a monotone interval pointer computes nearest distances and owners in $O(Q + K)$ time. All arrays are input- or support-sized. □

Radix sorting and floating-point key transforms are established techniques [3, 6]. This assumption does not give linear sorting of arbitrary real scores or a bit-complexity bound for growing precision. Conversion defines numerical ties, including collisions between distinct large integer inputs.

The same support-event width kernel combined with this preparation gives $O(n + (L + 1)Q)$ time. This is the basic Sparse VUS evaluator; its improvement over a dense rank sweep depends on Q, rather than recall saturation. The released PR-only implementation retains $L + 1$ scalar width results, adding $O(L)$ words; streaming removes this storage. Input-sized ranking arrays remain necessary in this construction, so the memory is not support-only.

3 PRIOR ALGORITHMS AND COMPARISON SCOPE

Boniol et al.'s optimized VUS algorithms retain threshold loops within widths and support-region summation [2]. The official source additionally creates full-length prediction and label arrays per threshold. Klein removes repeated threshold-wise summation through support cumsums, but still constructs maximum-support and full-length coverage arrays at each width [7]. Benchopt cumsums over all positions and computes component-hit counts through segment maxima [1]. Table 1 distinguishes these mechanisms from the new reductions. Its costs describe inspected operations, not cross-library runtime predictions.

Sorted sweeps, complete ties and trapezoids are established ROC techniques [4]. scikit-learn 1.0.2 explicitly truncates PR at full recall after forming cumulative counts [17]. Therefore full-recall stopping and zero-area PR suffixes are not new principles. Our earlier artifact comparison includes last-original-positive stopping, which already ensures saturation. The specific questions are whether buffer mass permits an earlier stop and whether preparation avoids paying for the discarded suffix. Reversible list updates [8], union-find [20], radix ordering [3, 6] and exact accumulators [13] are established tools used in the construction.

Sylligardos reports preliminary CPU/GPU VUS acceleration [18]. The public FF-VUS code attributed to him in aVUSi [19] uses a different ramp/overlap contract; its wrapper rounds scores, and Torch can interpolate component existence across widths when memory is insufficient. This is relevant acceleration prior art, but its native runtime is not an exact-target comparison. No result here establishes superiority over an unmeasured GPU evaluator. Source pins and minimal semantic witnesses are retained with the artifact; a package name alone does not establish equality.

Input: binary labels y, finite score matrix $S \in \mathbb{R}^{M \times n}$, integer maximum width L.

1. Validate inputs; enumerate original intervals and maximum support. Prepare two distances and original-positive indicators.
2. For each model, rank complete global ties. Retain each support position's event identifier and each event's complete global prediction count.
3. For $w = 0, \dots, L$, form support weights and merged-component identifiers once. For each model, accumulate grouped positive/buffer masses and first hits; scan its events with the exact PR identity and add the width integral.
4. Divide each accumulated integral by $L + 1$. Return values in input-model order.

Figure 1: Width-major shared PR evaluation. Model-specific threshold counts are never replaced by support-only counts. The algorithm stores one width's geometry, not an $(L+1)\times Q$ weight matrix.

4 SHARED-LABEL COLLECTION EVALUATION

Consider M score vectors with the same labels and width range. Independent calls repeat original-interval extraction, support enumeration, distance preparation, and width geometry. The *geometry* variant prepares the label-only arrays once and invokes an otherwise independent support kernel for each model. The *stream* variant additionally interchanges the width and model loops, so each width's support weights and component identifiers are computed once. It still builds model-specific grouped masses and integrates all complete support events. This is ordinary work sharing over the exact representation; it does not improve the asymptotic bound per model.

Proposition 2 (Shared evaluation). *For $M \ge 1$, the shared-label PR evaluator takes $O(Mn + (L + 1)MQ)$ time under Proposition 1's model. After validation and conversion, its retained evaluation workspace is $O(MQ+n+Q+K)$. The implemented public call has the more conservative $O(Mn+MQ+n+Q+K)$ peak auxiliary-space bound for its accepted inputs.*

Proof. Label-only preparation costs $O(n + Q + K)$. Ranking and complete global counts cost $O(Mn)$ and leave two model-by-support rank/count arrays. At one width, geometry costs $O(Q+K)$ and the M grouped scans cost $O(M(Q + K))$. Since $K \le Q$, summing widths gives the time bound. Ranking needs input-sized temporary arrays; the width kernel reuses support- and interval-sized arrays. This gives the retained bound. The public NumPy finite-value check materializes an $M \times n$ Boolean temporary, and conversion of other accepted dtypes can create an $M \times n$ float64 array. Including these yields the public-call bound. □

Construction	Threshold and width work	Semantic/comparison qualification
Boniol Algorithms 2/3	Per-width threshold loop; Q-support sums per threshold	Static-support reuse and cached predictions are prior techniques; the released target uses $\geq$ ties.
Klein fast-VUS	Per-width support cumsum, full-series coverage and threshold arrays	PR only; use $H = n$, not the sampled default $H = 250$. Explicit arrays cost $\Omega(Q + n + H)$ per width.
Benchopt	Per-width full-series cumsum and threshold arrays	Computes both ROC and PR; $H = n$ covers exhaustive ties. Explicit arrays cost $\Omega(n + H)$ per width.
FF-VUS	Batched width/threshold matrices and support masking	Native linear/max ramps differ from the pinned target; public wrapper rounds scores; Torch has an existence approximation fallback.
Sparse support events	$O(n + (L + 1)Q)$ with fixed-key preparation	Removes inactive thresholds while retaining global counts and exact ROC rectangles; no prefix stop is required.

Table 1: Algorithmic comparison. H is the requested threshold count; Q is maximum support. External array-cost lower bounds omit additional ramp, sorting and search work. Batched FF-VUS has no single sequential per-width loop. Our bounds use the stated arithmetic model.

This distinction prevents a misleading support-only memory claim. Score data already resident in float64 occupies $8Mn$ bytes independently of workspace. The stream variant's two int64 model-by-support arrays alone occupy $16MQ$ bytes; geometry-only sharing retains one model's arrays at a time. Thus sharing more work can require more live memory. We report process memory separately; NumPy validation temporaries cannot be inferred from the kernel's retained state.

Conditional alternatives. The artifact also implements reversible activation with incremental interval unions and stopping after complete-tie recall saturation. Its prefix bound is $O(n + L + \alpha(K)\sum_w J_w)$, where J_w counts the visited active positions. The stopping condition is $A + B/2 \geq P_0$ and $H = K_w$; both sides of the saturation predicate are monotone as predictions accumulate. For one original interval, a separate run aggregation implementation charges temporal runs of visited complete score classes. The detailed constructions and their versioned proofs remain in the artifact's earlier manuscript. These are conditional extensions, not the main speed claim of this paper. In particular, stopping at full recall, reversible lists, union-find and radix ordering are established tools [3, 6, 8, 17, 20].

5 EXPERIMENTAL DESIGN

Questions and frozen decisions. We ask whether support compression helps beyond a matched compiled dense implementation, whether shared labels improve the released sparse evaluator, and whether exact computation remains worthwhile against public exact and sampled APIs. A six-series development study compared three directions: direct complete-score evaluation, shared-label computation, and integer certification followed by exact fallback. Streaming improved over released independent calls by a geometric factor of 1.193. Certificate-plus-fallback took 33.93 times as long and certified no selection early. We therefore selected streaming before generating the new evaluation scores and retained the certificate route as a negative result. The primary width is 30; width 100 is secondary and never replaces an unfavorable width-30 result.

Data and score generation. The primary protocol selects all eligible NAB, SMAP and SMD files from the public TSB-AD-U archive [11]. It excludes six named development files, source-labelled synthetic files and exact observation-plus-label duplicates within the panel. The remaining 73 complete series comprise 20 NAB, 17 SMAP and 36 SMD cases, with lengths 1,000–28,696. The raw archive SHA-256 is `0c47020d`…; the primary protocol SHA-256 is `307e0498`…. Full digests, exclusions and decoded identities are part of the artifact. No series is silently replaced after scoring.

Eight fixed CPU procedures generate Isolation Forest, LOF, nearest-neighbor, one-class SVM, PCA, KMeans, HBOS and shrinkage-Mahalanobis scores. Each uses trailing length-16 windows with repeated boundary values and an at-most-2,048 point, evenly spaced, unlabeled fit set; random procedures use seed 20261024. This is an offline transductive workload for the evaluator. In particular, LOF novelty-mode scoring includes fit points; it is not a validated detector- accuracy protocol or a reproduction of eight tuned benchmark pipelines. All 584 stored outputs retain their float64 values without rounding or jitter. Of these, 569 have more than 101 distinct scores; the range is 3–28,696 and none is constant. One KMeans fit reports five distinct clusters for the requested 16; its warning and scores are retained. Unrounded does not mean tie-free.

Reuse and dependence. Sixty-two series were measured by name in previous repository studies. New score generation does not make the recordings unseen. Moreover, the 36 SMD series contain only nine distinct int8 label sequences; 18 share a label sequence with an excluded development file. Name exclusion is therefore not geometry-disjoint evaluation. We retain the frozen analysis and add two sensitivity analyses: equal weight per identical-label block, and exclusion of the development-matching geometry. Neither grouping proves independence of physical recordings, including normalized or shifted copies. The intervals characterize this workload, not an unknown population of future domains.

Methods and unchanged public baselines. The primary PR-only experiment compares stream, geometry-only, released sparse radix, active-prefix saturation, matched dense radix, and unchanged Klein fast-VUS at `thre=n`. The dense control uses the same compiled arithmetic and nearest-interval preparation; it isolates support compression from the choice of language/compiler.

After primary results, a separate protocol freezes comparison to unchanged Benchopt at `thre=n`, its matched two-output sparse-radix control, and Klein at 250 and 1,000 thresholds. Benchopt returns both ROC and PR. Its available cost to a PR consumer is useful, but only its comparison to the two-output control matches returned work. Klein is the faster measured external exact PR option on these collections. The source audit checks the current public revisions on 24 October 2026: Klein `1bae859f`, Benchopt `2b55ccb1`, official VUS `467bace0`, and aVUSi `facf4362`. FF-VUS's different native metric and unmeasured GPU execution remain outside same-output speed ratios.

Measurement and uncertainty. All numerical libraries use one thread. Each case evaluates all eight score vectors in one complete call, including validation, preparation, sorting, allocation and numerical winner selection. Twelve method orders consist of rotations and their reversals, balancing positions. All methods receive a tiny warm-up; imports and JIT compilation are outside the warm timing. Separate collection passes directly time reading every stored file through evaluation, selection, and decision-JSON write/close. They exclude detector fitting, cold-storage latency and `fsync`. A sum of case medians is never reported as a measured collection latency.

For each series, we take the median complete-call time over the twelve trials and form a baseline/candidate log ratio. We bootstrap these paired series ratios 10,000 times with seed 20261024. The primary tables report a three- collection Bonferroni 95% interval for each comparator; this is not a joint correction over every method, width and exploratory analysis. Workflow intervals resample explicitly paired trials on the fixed collection; trials are not independent datasets. Equal-label-block sensitivity averages log ratios within blocks before resampling blocks. Raw orders and unfavorable tails remain available. Balanced positions do not guarantee removal of all carryover or shared-host effects, especially for differences of a few percent.

Correctness and selection. All exact numerical calls must agree within 10^{-11}. Numerical winner sets include every value within 10^{-10} of the maximum, preventing an arbitrary index from resolving a near tie. For an approximate winner set, regret is the best exact value minus each selected model's exact value; we report both its minimum and maximum, with 0.01 fixed as a predeclared metric-scale cutoff. Approximate errors are retained rather than treated as reasons to exclude cases. The mathematical identity, floating-point tolerance, and exact rational certificates are distinct claims. The experiment does not certify all real-arithmetic rounding errors.

6 RESULTS

Whole method and sharing. Table 2 gives the frozen width-30 result. Stream is faster than released independent calls on every one of the 73 cases, with geometric ratios 1.496, 1.271 and 1.207 across NAB, SMAP and SMD. The smallest individual ratio is 1.132. The compiled dense comparison is 3.54–3.96 times slower, so the large public-library advantage is not solely an interpreter comparison. Removing the batch increment, matched independent-call dense/released ratios are 2.368, 2.905 and 3.279, with all 73 cases favoring support compression. The maximum numerical PR discrepancy over primary and supplementary exact calls is $1.11 \cdot 10^{-15}$, with identical numerical winner sets. The matched workflow ROC comparison differs by at most $4.86 \cdot 10^{-14}$.

Table 3 measures complete collection operations. Against unchanged exact Klein, the primary workflow ratios are 22.16–23.03. Against the released sparse precursor, they are 1.169–1.332, isolating the smaller sharing increment from the whole-method gain. Stream's three collection median latencies sum to 0.382 seconds; this sum is descriptive, not a directly timed joint-corpus query.

Where the new increment comes from. Geometry-only sharing already accounts for most of the SMAP/SMD increment. Stream/geometry improvement ratios are 1.094, 1.024 and 1.025, with stream losing on 3/17 SMAP and 2/36 SMD cases at width 30. Thus the new width-major loop is a modest refinement, especially on these collections. Its strongest reason for inclusion is the explicit decomposition of repeated work; the paper's algorithmic result remains the support-event representation. At width 100, released/stream ratios are 1.413, 1.255 and 1.221. Stream loses to saturation on four cases (two NAB, one SMAP, one SMD), with worst baseline/stream ratio 0.919. The separate earlier active-prefix study likewise found that saturation failed its default-width runtime criterion in all three runs. Those losses are consistent with extra scheduling overhead when little of the support can be skipped.

Dependence sensitivity. Equal weighting of the nine SMD label blocks gives released/stream 1.217, unadjusted 95% interval [1.191,1.246], and Klein/stream 23.76 [20.68,26.59]. Removing the 18 cases whose label geometry also appears in development leaves 18 cases in eight blocks: released/stream is 1.220 [1.191,1.253]. These analyses preserve the direction of the improvement while narrowing the inference. They do not turn reused recordings into a new-domain test.

Strong public comparators and approximation. The separately frozen study corroborates Klein as faster than Benchopt's available exact API on all three collections. Comparing equal ROC+PR outputs, the sparse-radix implementation improves directly timed workflow latency by 25.3, 32.1 and 40.9 times (Table 4); all paired trial intervals exclude one. This comparison does not attribute the whole effect to the new PR batch loop.

Klein at 250 thresholds changes 14 of 73 numerical winner sets, all in SMAP or SMD; four cases have minimum and maximum regret at least 0.01. The largest regret is 0.0584. At 1,000 thresholds, three winner sets change, with no regret at least 0.01 and maximum 0.00346. NAB has zero winner changes at either resolution. Nevertheless, complete exact stream workflows are faster than 250-threshold Klein by 11.25, 9.68 and 7.27 times. On this workload, sparse exact evaluation improves both numerical fidelity and measured runtime. The result does not imply that approximation always selects a worse model, that VUS is the correct application utility, or that our detector procedures are superior.

Observed support regimes. At width 30, maximum-support fractions range from 0.0759 to 0.223 in NAB, 0.00799 to 0.140 in SMAP, and 0.00384 to 0.1148 in SMD. Original component counts

Collection	Cases	Released / stream	Dense / stream	Klein exact / stream
NAB	20	1.496 [1.428, 1.566]	3.543 [3.338, 3.713]	30.513 [27.399, 33.652]
SMAP	17	1.271 [1.242, 1.302]	3.693 [3.196, 4.191]	25.772 [22.941, 28.940]
SMD	36	1.207 [1.193, 1.221]	3.956 [3.736, 4.125]	25.172 [23.475, 26.695]

Table 2: Primary complete-call geometric speedups at width 30. Brackets give paired-series 95% bounds adjusted across three collections for each comparator. All include public validation, preparation, allocation and eight-model winner selection. Hardware is identified by the selected evidence receipt; per-series intervals are conditional on the fixed panel and supplemented with label-block sensitivity.

Collection	Stream	Released	Dense	Klein
NAB	0.0470	0.0626	0.1565	1.0828
SMAP	0.0528	0.0647	0.1756	1.1699
SMD	0.2824	0.3302	1.0328	6.3759

Table 3: Directly timed primary collection workflow medians, seconds. Each pass reads all score files and writes/closes its selected-model output. Detector fitting, cold storage and JIT compilation are excluded.

range from 1 to 5, 1 to 3, and 1 to 12 respectively. These are genuinely sparse supports, the regime predicted to benefit. When $Q \approx n$, the support-sensitive asymptotic benefit disappears; extra indexing and batch storage can dominate. Short cases also make small sharing differences sensitive to allocation and interpreter overhead. Resource stress cases examine these regimes separately and do not enlarge the count of real datasets.

Does a one-expression change explain the gain? The PR inactive-event corollary admits a simple change to Klein: replace its global threshold vector by the distinct scores on maximum-width support, keeping global predicted-positive counts and every other operation unchanged. We freeze this author-created ablation after the preceding studies and before its own measurements. It reuses all 73 series and eight stored scores, with eight balanced case orders and three not-fully-balanced workflow passes. All scalar values and numerical winner sets agree; maximum PR discrepancy is $1.11 \cdot 10^{-15}$.

The patch improves Klein by 1.654, 2.274 and 2.968 times, confirming that the elementary threshold restriction explains part of the public-library gap (Table 8). The released single-model evaluator remains 12.052, 9.067 and 6.969 times faster than this patch, with all 73 case medians and each adjusted interval favoring it. For SMD, equal-label-block weighting gives 6.772 [5.835,7.253], with the same three-collection correction. The patched/stream workflow median ratios are 11.96, 10.14 and 7.46. These residual ratios include NumPy-versus-compiled implementation effects; the matched compiled dense control separately isolates support compression. The patch still performs full-length coverage/segment-max work per width and does not supply the ROC correction. Accordingly, we claim the complete exact operator representation and elimination of repeated width-dependent work, not that PR threshold restriction alone is a deep new primitive.

Compilation-matched conventional support control. A second, post-review control uses the same validation, binary64 radix ordering and complete global tie counts as our implementation, but reconstructs each width with direct sum-and-clip ramps and component support scans. It caches original-positive group mass across widths and uses support-local scratch. This is an optimized author reimplementation of the support-cumsum approach, not an unchanged public release or a literal compilation of Klein. It shares no nearest-distance geometry. Six balanced orders measure all 73 fixed cases and six complete workflow passes per collection; maximum PR discrepancy is $4.44 \cdot 10^{-16}$.

The result limits attribution (Table 9). Conventional control/released case ratios are 0.910, 0.960 and 0.978, with every adjusted interval below one and the control faster on all 73 cases. Thus this panel does not establish a separate runtime advantage for nearest geometry over an optimized compiled direct-ramp method. Control/stream ratios are 1.357, 1.201 and 1.166, and complete-workflow ratios are 1.177, 1.126 and 1.152. These remaining gains include shared preparation and width work. They are much smaller than the public NumPy-library comparisons. The direct-ramp control does not refute the exact support-event identity or the stated bound, but it precludes attributing the public-library gap to that geometric lemma alone.

A public multi-metric consumer and its bottleneck. To test an existing consumer rather than only our VUS-specific selection loop, we pin TSB-AD's unchanged `get_metrics` function at revision `303dffbf` [11]. Its scalar VUS callback receives either exact Benchopt ROC+PR or our exact radix pair; all seven other metric outputs and their upstream oracle-threshold searches remain unchanged. The recording is the largest primary input by length (SMD id 5, $n = 28{,}696$), selected before consumer outcomes. Its eight stored profiles are evaluated in four complete read-to-CSV-close passes in ABBA order, two per backend. The unchanged public autocorrelation rule selects width 125. This is a stored-profile consumer, not the upstream detector-training script or a deployed service. Source hashes are checked against the pinned revision even on cache hits.

The trace finds a strong Amdahl limit. Median VUS time falls from 0.9289 to 0.01406 seconds (66.07×), but the complete nine-metric workflow falls only from 34.260 to 33.234 seconds (1.031×). VUS accounts for 2.70–2.72% of baseline time; the remaining metric work dominates this consumer. All nine outputs agree to $3.45 \cdot 10^{-15}$. Two fixed-input repetitions per backend provide descriptive timing, not a population interval. The large VUS-specific workflow ratios therefore must not be advertised as a similar end-to-end speedup for this multi-metric benchmark.

Collection	Paired case Benchopt / radix	Benchopt workflow (s)	Radix workflow (s)	Workflow ratio
NAB	27.284 [25.680, 28.780]	1.9961	0.0788	25.32
SMAP	34.567 [31.285, 38.479]	2.2664	0.0707	32.06
SMD	44.160 [41.109, 47.075]	14.5331	0.3551	40.92

Table 4: Supplementary matched-output comparison: both methods return exact numerical ROC and PR. Width 30, unchanged Benchopt source, same 73 series and scores. Case bounds use the same three-collection adjustment. This gain belongs to the complete sparse evaluator, not solely to the new PR batch loop.

	Klein, 250 thresholds			Klein, 1,000 thresholds		
Collection	Changed sets	Regret ≥ 0.01	Max regret	Changed sets	Max regret	250 / exact workflow
NAB	0	0	0.00000	0	0.00000	11.25
SMAP	6	3	0.05835	1	0.00345	9.68
SMD	8	1	0.05431	2	0.00283	7.27

Table 5: Approximation accuracy–cost comparison. Minimum and maximum regret threshold counts coincide in this panel; both are retained in the artifact. The final column compares directly timed PR-only workflows. Zero losses, including all NAB selections and every 1,000-threshold loss below 0.01, are retained. Values describe these fixed detector-score inputs, not detector quality.

Positive fraction	Released	Geometry	Stream	Dense
0.01	181.0	183.2	198.4	217.6
0.50	175.2	188.1	225.0	210.6

Table 6: Median OS lifetime peak RSS (MiB), three fresh processes per method, $n = 500{,}000$, $M = 8$, one centered anomaly interval, width 30. These synthetic resource cases include interpreter, libraries, inputs and warm-up; they are not measurements of auxiliary space or additional real datasets.

Long recordings and controlled operating range. A follow-up frozen after the primary study and review, but before its own score generation and timings, includes every MITDB and SWaT recording in the same public archive: eight MITDB series and one SWaT series, length 419,919–650,000, with 2–199 original components. Eight names appeared in older repository studies; MITDB has seven distinct label sequences. These are additional fixed workloads, not nine independent unseen populations. All use the same eight detector procedures, width 30 and seven cyclic method orders. Separate collection workflows use only three orders and are not fully position-balanced. Exact-output PR and ROC discrepancies are at most $2.56 \cdot 10^{-15}$ and $8.62 \cdot 10^{-14}$, respectively.

MITDB and SWaT workflow median ratios for Klein/stream are 23.10 and 17.70; MITDB's latency drops from 53.14 to 2.30 seconds (Table 7). Matched ROC+PR ratios are 29.50 and 25.66. The additional sharing gain is less decisive: released/stream workflow median ratios are 1.220 and 1.086, but paired-geometric trial intervals [0.925,1.340] and [0.955,1.165] include one. Stream also loses to geometry-only sharing on one MITDB case. Thus long-recording evidence supports the whole evaluator more strongly than the small batch increment.

Sixteen deduplicated synthetic conditions vary one axis from a common $n = 100{,}000$, $M = 8$, positive fraction 0.01, $K = 1$, width-30 anchor: n up to one million, positive fraction up to 0.9, K up to 1,000, M up to 32, width up to 300, and a 16-level tied-score condition. Four cyclic orders balance method positions. These cases test evaluator cost, not detector accuracy. Figure 2 reports paired trial geometric ratios, with unadjusted conditional 95% intervals. At positive fraction 0.9, matched dense/released is 0.980 [0.962,0.9994]: support compression no longer improves runtime (the ratio of medians is 0.995). At width 0 its ratio is 1.065, versus 16.36 at width 300. With $M = 1$, released/stream is only 1.023; sharing has little work to amortize. These observations delimit the contribution rather than imply universal dominance. Increasing K also increases expanded support and is not an isolated intervention on component overhead.

Post-timing diagnostics separately instrument the earlier active-prefix algorithm. A dense independent reconstruction of interval ramps, complete score ties and saturation verifies every visited-count array. At the one-million-point random-score case, this algorithm visits 99.94% of active positions, explaining why recall saturation is not a general substitute for support compression. At $K = 1{,}000$, it visits 33.79%; these counts do not measure the different stream implementation's work or establish its timing.

Fresh execution on a second CPU.. The one-command artifact was executed from a fresh checkout on a hosted Linux runner with an AMD EPYC 7763 CPU, using the same source revisions, cohort, detector parameters and balanced method orders. Raw data and public comparator sources were downloaded and all scores generated anew. Both independent record audits passed. At width 30, released/stream case ratios were 1.886, 1.580 and 1.340 for NAB, SMAP and SMD; the respective three-collection-adjusted 95% intervals were [1.732,2.043], [1.526,1.644] and [1.322,1.359]. All 73 case medians favored stream. The exact Klein workflow ratios were 20.06, 20.28 and 16.17, while matched ROC+PR Benchopt/radix workflow ratios were 22.38, 30.40 and 42.55. Thus the direction of the main result survives a second CPU and a fresh

Collection	Series	Stream (s)	Klein (s)	Released (s)	Paired geometric Klein/stream [95% CI]	Matched pair ratio
MITDB	8	2.300	53.141	2.806	22.10 [19.00, 25.00]	29.50
SWaT	1	0.245	4.339	0.266	17.62 [16.37, 18.90]	25.66

Table 7: Separately frozen long-recording follow-up, $M = 8$, width 30, Apple M1 Pro. Times are medians of three complete collection passes. PR intervals resample paired timing trials on fixed inputs and are unadjusted; three workflow orders are not fully position-balanced. The matched-pair column is the ratio of median Benchopt and radix workflow times for ROC+PR. SWaT is one recording, not an estimate over a population of independent series.

Collection	Klein / patched Klein	Patched Klein / released	Patched Klein / stream	Workflow patched/stream
NAB	1.654 [1.496, 1.842]	12.052 [10.325, 13.812]	17.956 [14.605, 21.626]	11.96
SMAP	2.274 [2.149, 2.401]	9.067 [8.179, 10.052]	11.534 [10.190, 12.994]	10.14
SMD	2.968 [2.842, 3.089]	6.969 [6.586, 7.270]	8.396 [7.891, 8.794]	7.46

Table 8: Author-created minimal ablation: replace only Klein's threshold list with all distinct maximum-support scores; retain global counts and other code. Eight balanced case orders on the original 73 series, width 30, Apple M1 Pro; case intervals are 95% bounds adjusted across three collections per comparison. The final column uses three not-fully-balanced workflow passes. Remaining gains include implementation differences; the compiled dense control separately isolates support compression. The patched code is not an upstream release.

Collection	Control / released	Control / stream	Released / stream	Workflow control/stream
NAB	0.910 [0.905, 0.915]	1.357 [1.291, 1.427]	1.492 [1.419, 1.567]	1.177
SMAP	0.960 [0.945, 0.973]	1.201 [1.180, 1.224]	1.251 [1.218, 1.287]	1.126
SMD	0.978 [0.969, 0.985]	1.166 [1.158, 1.174]	1.193 [1.181, 1.205]	1.152

Table 9: New author-created compiled conventional support-cumsum control, with matched validation, radix ordering and global counts, direct per-width ramps and cached original-positive mass. Six balanced orders for all 73 primary cases and six balanced workflow passes, width 30, Apple M1 Pro. Brackets are three-collection-adjusted 95% intervals. Ratios below one favor the control; all scalar cases favor it over released. This is not unchanged public Klein.

input-generation run. Hosted-runner load is uncontrolled: one NAB stream workflow took 0.111 seconds against a 0.061-second median and is retained. We report each host separately rather than pooling its timing repetitions.

Fresh detector scores differ across hosts, and so does the approximation result: 250 thresholds change 13/73 winner sets on Linux, including three losses of at least 0.01, versus 14/73 and four on the Mac. The difference is one SMAP case whose approximate selected model changes, while its exact winner remains the same. Its SVM exact VUS-PR changes from 0.3093 to 0.3329. A post-review diagnostic archives fresh Mac/Linux score arrays, standardized inputs, fitted SVM state and numerical-backend metadata. Raw inputs and standardized matrices are byte-identical, but the fitted support-vector counts are 104 and 105. Both arrays reproduce the two SVM metric values on one evaluator host and agree with unchanged Klein to $5.56 \cdot 10^{-17}$. This localizes the difference after standardization to fitted-state/score generation; it does not identify a particular compiler or numerical instruction as causal. The new Linux eight-model score hash differs from the first runner (other metric differences reach 0.000583), so these arrays are a diagnostic repeat, not recovered original-run arrays. The Mac score hash exactly matches its first run. The 1,000-threshold counts remain three changed sets and zero losses of 0.01. This sensitivity reinforces the distinction between a fixed-input numerical comparison and a cross-platform detector-selection claim.

Process memory. A separately frozen resource study selects the real cases with largest length, support fraction and component count, repeats the largest case's eight scores four times for a declared $M = 32$ stress case, and adds two synthetic density cases at $n = 500{,}000$, $M = 8$. Three fresh processes per method record OS lifetime peak RSS, including interpreter, libraries, inputs and tiny JIT warm-up. This measures process resource use; subtracting two high-water marks does not measure exact auxiliary space. On the real $M = 8$ cases, stream peaks at 113.9–119.0 MiB; on the repeated $M = 32$ case, 127.4 MiB versus released 125.5 MiB. The two synthetic cases expose the tradeoff more clearly (Table 6). At positive fraction 0.5, stream's peak is 225.0 MiB versus 175.2 MiB for released and 210.6 MiB for dense. Thus the batch speed claim does not imply lowest peak memory. The largest numerical PR discrepancy in this resource study is $1.67 \cdot 10^{-16}$; no timing conclusion or real-data generalization is drawn from its synthetic inputs.

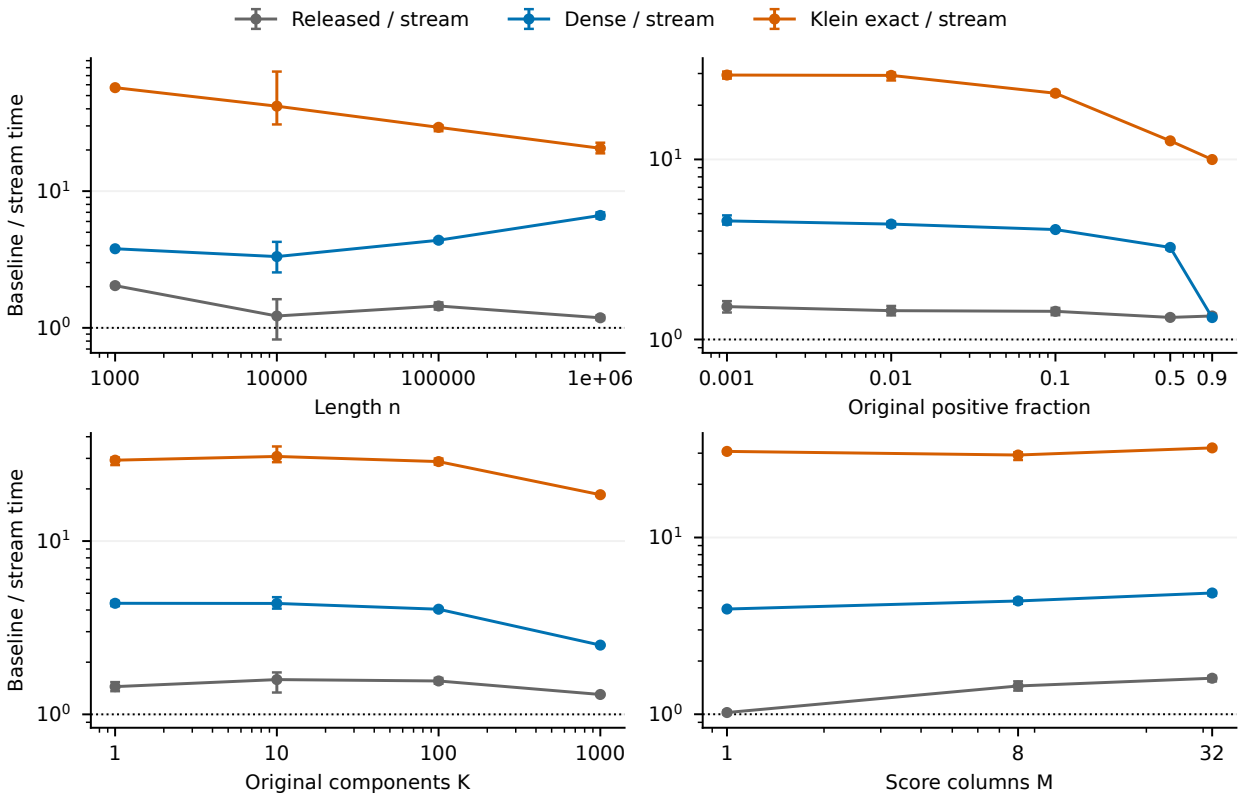


Figure 2: Controlled one-axis runtime conditions. Points and bars are paired geometric time ratios and unadjusted 95% trial intervals on four fixed-input orders. Above one favors stream. All other parameters retain the common anchor; the component axis also changes expanded support. These synthetic mechanism checks supplement, rather than enlarge, the real-recording panel.

7 RELATED WORK AND NOVELTY BOUNDARY

The closest algorithmic work is the original VUS operator and its optimized variants [2, 15]. We preserve that operator. Temporal weighting and proximity-aware PR are also established by PATE [5]; we neither introduce those ideas nor compare a different metric's runtime as if it returned VUS. Sylligardos's efficient metric work [18] and current public implementations motivate the semantic and source audit in Section 3.

TSB-AD [11] studies large detector benchmarks and metric reliability; TAB [16] unifies benchmark data, methods and evaluation. Our operator is a possible component of those systems, not a replacement benchmark. TSB-AutoAD [10] compares selection, ensembling and generation; RAMSAD [14] retrieves historical detector profiles for selection. We reduce the cost of computing labeled profiles, without claiming superior unlabeled selection. mTSBench [25] covers 344 multivariate series and 24 detectors, broader than our targeted univariate operator workload. Analyses of foundation-model pipelines [26] likewise address detection methodology rather than the exhaustive scalar representation studied here.

Recent formal analysis of 37 metrics explicitly excludes score-based volume-under-surface metrics from its formal scope [22]. The resulting LARM/ALARM axioms do not establish our computational identity. CCE [24] gives a different linear-time metric; its speed is not an equal-output baseline. The metric taxonomy [23] concerns application requirements. Adversarial seed-shopping analysis [12] varies reporting choices and compares metrics, while our approximation experiment holds labels, score vectors and the target fixed. None of these distinctions makes rank changes themselves a new discovery. The narrow contribution is the exact event reduction, complete global counts, dynamic integral correction and its measured implementation.

8 REPRODUCIBILITY AND SCOPE

The artifact binds the public archive, each raw member, decoded observations, labels, scores, method sources, trial orders and output records by SHA-256. A single command creates a new isolated workspace, downloads pinned sources, generates scores, executes both studies and reconstructs their numerical tables as JSON (TeX rendering and manuscript compilation are separate):

```
python experiments/reproduce_pvldb20261024.py
  --output /tmp/pvldb-replica
```

It refuses an existing destination. Independent reconstruction imports no metric producer and checks complete coverage, input/protocol/run bindings, scalar values, winner sets, paired intervals and every principal-table field. Corruption tests exercise changed intervals, fabricated summary entries, replaced inputs, duplicate cases and false regret. Historical audit revisions remain in the artifact. Fresh execution on the second CPU additionally checks the full pipeline beyond reconstruction of recorded outputs.

The recorded environment is Python 3.12.4, NumPy 1.26.4, Numba 0.59.1, pandas 3.0.1, scikit-learn 1.8.0 and SciPy 1.17.1. The public consumer uses a separate statsmodels 0.14.6 overlay; an earlier 0.14.2/pandas import failure is retained and occurred before any measurements. Detector scores can differ across platforms even with these package pins; the diagnosed SVM fitted states differ despite identical standardized inputs. Each execution workspace retains its CPU, platform, package versions and independently generated arrays. The archived Linux output contains records and input hashes but omits the generated NPZ arrays; a later inspectable two-host diagnostic pair is archived separately. We do not pool hosts into a single interval. Our CPU measurements do not establish GPU superiority; the literature comparison is a bounded search of the stated sources.

The principal one-command execution covers the primary and public-comparator supplement only. Long-recording/controlled, resource, minimal-Klein and compiled conventional-support and public-consumer studies have separate frozen protocols and commands in the artifact instructions. The operating-range wrapper regenerates its inputs and checks its full design before timing; its stages were executed in the recorded run, while the wrapper itself has not received a second complete fresh execution. This boundary is separate from the completed fresh primary/supplement run on Linux.

Limits. The principal evaluation uses short-to-moderate univariate series with sparse anomalies and shared labels; the separate follow-up reaches 650,000 points on real data and one million synthetically. Dense support and numerous fragments are covered by controlled cases, not broad real-domain samples. Different component-merging conventions, other VUS ramps, cold compilation and training-dominated pipelines remain separate regimes. The conventional compiled control is faster than our released single-model method at the primary width, so separate geometric superiority is not claimed. Whole-operator speed can matter for repeated stored-profile queries even when the absolute savings on one small collection are seconds. That use case must not be generalized to end-to-end detector training. The mathematical statements use unit-cost metric arithmetic; finite-precision experiments provide numerical parity, not an all-input formal rounding proof. Batching uses more retained support state, and its full public peak includes model-by-time validation/conversion arrays.

9 CONCLUSION

For the original dynamic VUS operator, inactive thresholds can be removed from an exhaustive scalar computation only with complete global tie counts and the appropriate ROC predecessor term. The resulting support-sensitive evaluator preserves the target and permits shared-label evaluation. Three public collections show numerical agreement and lower complete-call and workflow cost than the measured exact comparators; the advantage also holds against the measured sampled API. Modest incremental sharing gains, unfavorable prefix cases, dependence and memory costs define the supported scope.

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